

REDEM-SSM: A State-Space Architecture with Native Online Learning, Meta-Adaptation, and Structural Plasticity

Yaming Hu^{a,*}

^a*Independent Researcher, Guiyang, Guizhou, China*

Abstract

State-space models (SSMs) are a natural host for online learning under domain shift, but it is unclear which adaptation mechanisms must be native to the host rather than retrofitted. We instantiate four REDEM mechanisms — an online readout (M1), a slow-trace statistical memory (M3), soft routing over trained specialist readouts (M4, this host’s instantiation of the framework’s structural-plasticity slot), and a state-norm homeostat (M5, this host’s instantiation of the stability-regulation slot) — on a diagonal linear recurrence with a log-uniform timescale spectrum, following a falsifying 10-seed paired development sequence. Three host requirements emerge. (i) A linear readout cannot recover the current token from a linearly-decayed state mixture (a deconvolution impossibility for affine-linear exact recovery); the readout must consume the additive input path, which beats a Transformer+LoRA reference on every seed, and fixed multiplicative gates do not substitute for learned selectivity. (ii) The state’s role is statistical metadata: a fast-channel exponential moving average tracks domain switches and soft routing retains per-domain specialists. Under the clipped-linear perplexity of this squared-loss RLS readout family, several stream comparisons are floor-sensitive and reverse once clipped tokens are excluded; forgetting gains and the state-norm-homeostat negative result are robust. A companion study shows the artifact belongs to the training objective, not the output range: post-hoc simplex projection removes negative mass without repairing stream scores, whereas a trained probabilistic readout does. (iii) A state-norm homeostat restores full-state statistics on a two-

*Corresponding author. ORCID: 0009-0003-1406-0485.

Email address: 64687555@qq.com (Yaming Hu)

domain task but degrades on a four-domain benchmark and is excluded from the full stack. On that benchmark the full stack beats the bare host and the untuned Transformer+LoRA reference on every seed on forgetting (stream margins against the bare host are floor-sensitive). All results are CPU-scale proofs of concept; we make no scaling claims.

Keywords: online learning, state-space models, continual learning, reservoir computing, homeostatic regulation, timescale coverage

1. Introduction

Online sequence models must adapt while the stream is still running: domain switches, non-stationary statistics, and long-horizon context arrive without an offline retrain. State-space models (SSMs) replace the frozen feedforward stack of Transformers with a recurrence whose timescales can be designed rather than only learned [1, 2], and they inherit a structural kinship with reservoir computing: a fixed substrate, a trainable readout, and an explicit separation between dynamics and plasticity [3, 4]. The open design question is *which* adaptation mechanisms must live natively in such a host.

The REDEM framework couples four such mechanisms: a continuously updated online readout (M1), a slow exponential trace — an exponential moving average (EMA) of the fast channels — that supplies statistical memory (M3), structural plasticity (M4), and a homeostatic regulator that keeps the substrate at a stable operating point (M5) — the substrate-level regulation of the adaptation process itself is the framework’s *meta-adaptation* [5]. A numbering note before we use these labels: the companion papers number mechanisms per *function* (in B, M4 is coupling-graph rewiring and M5 a chaos homeostat), whereas this paper numbers per *host slot* and instantiates them differently — here M4 is soft routing over trained specialist readouts (a host-specific instantiation of the gradual-structural-change design principle of [6], not graph rewiring) and M5 a state-norm homeostat on the whitened-state statistics. We use the companion numbering throughout (M1, M3–M5) with that instantiation convention stated once here; M2, the reward-gated plasticity mechanism of [5], is not required by the RLS-only readout constraint adopted here and is not instantiated on this host. A companion study [6] tested the same mechanism set on a Transformer host with low-rank adapters: slow-trace domain routing transferred (forgetting -28% , 10/10 seeds) while a gating-only variant was falsified at every tested metadata timescale (0/10). That work

explicitly did *not* test a frozen-feedforward host and did *not* claim a general host boundary; the constructive reading it supports — a persistent stateful host rather than a retrofit onto a static architecture, for the state-carrying mechanisms that transferred there — is the reading this paper takes as a design premise and tests by building the host.

This paper takes the constructive consequence. We design a stateful host — a diagonal linear state-space recurrence with a log-uniform timescale spectrum (the timescale-coverage principle of [6]) — and instantiate M1, M3, M4, and M5 natively. Rather than asserting that the design works, we follow the falsifying discipline of the companion work: every claim is a paired 10-seed comparison with sign consistency, and the negative results are reported as design evidence, not as failures.

The development sequence produced three host requirements that are the paper’s main contribution:

1. **The readout needs a calibrated input path, not a clipped state mixture** (Section 4.1). Exact recovery of the current token from a linearly-decayed state mixture is a deconvolution impossibility (Proposition 4.1). Under the clipped-linear stream metric, every state-feature arm fails out of sample (0/10 vs. the Transformer+LoRA reference); that ranking is a *calibration* property, not an absence of linear information — coverage-spectrum state arms achieve lower unclipped CE while parking 10–12% of tokens on the clip floor (the log-normal arms park 7.5–8.3%), whereas the input path produces non-positive scores on only $\sim 1.6\%$ of tokens. Fixed Mamba-style gates do not repair calibration (both $\gamma \in \{1, 5\}$); selectivity must be learned (a boundary we flag, not cross).
2. **The state’s role is statistical metadata** (Section 4.2). A fast-channel state EMA (M3) tracks domain switches and routing over domain specialists (A3) retains them: forgetting improves at $\tau_m \leq 1000$ under both the reported and the unclipped metric. The pause-learning policy (A2) does *not* pass the paper’s own rubric: its reported stream gain over A1 reverses once A1’s clip floor is removed, and forgetting is never better than A1 at $\tau_m \in \{200, 500\}$ and is only weakly better at larger τ_m .
3. **Specialist retention beats a single stream readout, and state regulation is conditional** (Section 4.3). At the tested $\tau_m=500$ point, soft routing retains purer forgetting than a bare single readout, while

abrupt hard switching retains purer specialists than soft routing on forgetting; the reported soft-over-abrupt *stream* margin reverses under the unclipped metric (Section 4.7). A state-norm homeostat (M5) bounds the state and restores the full-state EMA as a valid domain statistic *on the two-domain task only*. On the four-domain benchmark M5 degrades both metrics and is excluded from the full stack (Section 4.4).

None of the constituent mechanisms is individually new — input-path features, softmax routing, fast-channel EMAs, and norm feedback are known primitives — so the contribution claimed here is the falsifying development sequence itself, the demonstration of which mechanism combination a stateful host needs natively, and the log-uniform timescale spectrum as the host-design choice. This is a host-design study in the reservoir/online-readout lineage [4, 7, 8], not a claim of a new sequence architecture.

On a four-domain irregular-switch benchmark the full stack beats the bare host and the Transformer+LoRA reference on every seed (the reference untuned for four domains; tuned, its stream margin narrows to -1.68 while its retention collapses, Section 4.4), and the result transfers to a two-book real-text protocol (Section 4.4). All experiments are CPU-only and tiny by design (scaling disclaimed in Section 7).

2. Related work

State-space sequence models and online adaptation.. State-space models replace the frozen feedforward stack of [9] with a recurrence whose parameters are trained end-to-end ([1, 2, 10, 11]); online adaptation has been pursued by rewriting the state with a learned optimizer ([12]) and by linear-attention delta updates ([13]). Those hosts train with backpropagation and typically assume a large pretrained parameter set, whereas the REDEM line ([14, 5, 6]) restricts the readout to an online, local rule on a fixed substrate. Our host therefore sits closer to the designable-timescale end of the SSM spectrum than to end-to-end selective SSMs such as Mamba.

Reservoir computing and fixed-substrate readouts.. The fixed-substrate, readout-only split is classical in reservoir computing: echo-state networks and liquid state machines run a fixed, randomly generated recurrent substrate and train only a linear readout ([4, 7, 8, 15], surveyed in [3]). Our host shares that split, but replaces the randomly wired reservoir with a diagonal linear recurrence whose log-uniform timescale spectrum is the design degree

of freedom that the readout and the metadata consume; nothing recurrent is trained, and the adaptivity lives entirely in the online readout and the routing mechanisms (Sections 3.2–3.4). Relative to classical ESNs, the paper’s contribution is not a new reservoir topology but a controlled account of which auxiliary mechanisms (input-path features, metadata EMAs, soft specialist routing, optional homeostat) the host needs under non-stationary streams.

Continual and online learning practice.. Continual-learning practice emphasizes stability–plasticity trade-offs and non-stationary evaluation ([16]); online convex optimization supplies the regret-theoretic frame for linear predictors updated per round ([17, 18]). Our evaluation protocol follows that spirit — paired per-seed stream and forgetting perplexity under known and irregular domain switches — but the scientific object is the *host requirement*, not a new regularizer or a new continual-learning algorithm class. Our Transformer+LoRA reference ([19], retrained per task as in [6]) is a tiny parameter-efficient adapter and is not compared against larger models (Section 7); a classical reservoir+RLS baseline is architecturally close to our bare host, so we additionally run a separately wired ESN+RLS control on the identical protocol and metrics (Section 4.6); it fails out of sample.

3. Methods

3.1. Host: diagonal state-space recurrence

The host is a diagonal linear recurrence

$$h_t = A h_{t-1} + B e_{t-1}, \quad A = \text{diag} \left(e^{-1/\tau_i} \right), \quad (1)$$

where e_{t-1} is the one-hot token at step $t - 1$, $B \in \mathbb{R}^{N \times V}$ is a fixed random input projection (entries drawn i.i.d. $\mathcal{N}(0, 1/N)$, per-seed), and the timescales τ_i are drawn log-uniformly over $[1, 3000]$ tokens ($N = 128$, $V = 32$). The log-uniform spectrum realizes the timescale-coverage principle of [6]: fast channels carry the recent token, slow channels carry long-horizon statistics. The state is spectrum-whitened,

$$h_t^w = h_t \odot \sqrt{N(1 - A^2)}, \quad (2)$$

which normalizes each channel to stationary unit variance (the steady-state scaling of the linear recurrence) and bounds slow-channel magnitude; without it the slow channels accumulate and the RLS covariance grows as $(1/\lambda)^t$ in unexcited directions.

3.2. M1: online RLS readout on the input path

The readout is a per-token recursive-least-squares linear map

$$\hat{y}_t = W \phi_t, \quad \phi_t = [Be_{t-1}; 1], \quad (3)$$

with $W \in \mathbb{R}^{V \times (N+1)}$ updated online (predict before update) against the one-hot target e_t with forgetting factor $\lambda = 0.9999$ — the classical recursive-least-squares family ([20, 21, 22]), also an instance of online convex optimization for linear predictors ([17, 18]) — and with the inverse covariance initialized as $P_0 = \delta I$ with $\delta = 1$. The cost is $O(F^2)$ per token in the feature dimension $F = N + 1$, not $O(P^2)$ in a parameter count [5, 6]. The squared-loss readout output is a linear minimum-mean-square-error (MMSE) estimate of the conditional distribution, so perplexity is evaluated as the cross-entropy $\text{CE} = -\ln \text{clip}(\hat{y}_t[e_t], 10^{-12}, 1)$ *without* a softmax (a softmax double-normalizes the estimate toward uniform). The fraction of non-positive scores, `neg_frac` ($\hat{y}_t[e_t] \leq 0$), is reported as a diagnostic. Because clipping removes the standard probabilistic meaning of perplexity, cross-arm comparisons inherit the clip range. We therefore report, for every SSM arm, the clip fraction `clip_frac` (tokens with $\hat{y}_t[e_t] \leq 10^{-12}$, the true floor hits of the CE definition above — in all measured arms this coincides with `neg_frac`, because scores in $(0, 10^{-12}]$ were never observed), the holdout counterparts `forget_neg_frac` / `forget_clip_frac`, and two unclipped diagnostics — `ppl_unclipped` (each arm on its own $\hat{y} > 0$ tokens) and a same-token-set rescore of every pair on the intersection of their $\hat{y} > 0$ supports (Section 4.7). Per-token arrays enabling the same-token-set rescore are written under `data/per_token/`. The unclipped metrics are *sensitivity analyses*, not a replacement headline: selection bias remains for the per-arm form, the same-token set still drops tokens on which either arm clips, and both are undefined for arms with no non-positive scores under this evaluation (the Transformer+LoRA reference uses softmax cross-entropy and has no clip floor). Section 4.7 tabulates which conclusions survive the floor removal.

The input-path feature choice is itself an evidence-driven result (Section 4.1); the whitened state h^w is *not* a readout feature.

3.3. M3: fast-channel metadata EMA

M3 is a per-token exponential moving average of the fast channels of the whitened state,

$$m_t = \left(1 - \frac{1}{\tau_m}\right) m_{t-1} + \frac{1}{\tau_m} h_t^w[\text{fast}], \quad \tau_i \leq 8 \text{ tokens}, \quad (4)$$

with $\tau_m \in \{200, 500, 1000, 2000\}$. The fast channels (which converge in a few tokens) are a stationary domain statistic; the full state is not — its slow channels accumulate over the whole stream, so an EMA of the full state drifts away from any fixed per-domain reference and the detector never flips. Fixed per-domain references are the mean fast-channel state over 1500-token reference streams generated per domain.

3.4. M4: soft routing over trained specialists

For D domains, M4 maintains D readouts (W_d, P_d) . Routing uses continuous softmax weights over the EMA distances,

$$w_d = \frac{\exp(-\|m_t - r_d\|/\kappa)}{\sum_{d'} \exp(-\|m_t - r_{d'}\|/\kappa)}, \quad \kappa = 0.5 \times \text{median pairwise } \|r_d - r_{d'}\|, \quad (5)$$

with the prediction $\hat{y}_t = \sum_d w_d W_d \phi_t$ and W_d, P_d updated with error scaled by w_d . Here κ is the soft-routing length scale (data-adaptive, of order the reference separation), distinct from the substrate coupling strength κ of the companion substrate paper [14] (there $\kappa^* \in (25, 30)$) and from the E/I modulation ratio κ of the silicon-nitride front-end; the three are unrelated quantities. This instantiates the gradual-structural-change design of [6]; on this host the reported soft-over-abrupt stream margin is floor-sensitive and is not claimed as primary evidence (Section 4.7). Continuous softmax routing over specialist readouts is the online, single-layer analogue of mixture-of-experts routing [23], and the gating-versus-gradual contrast echoes the gating practice of recurrent networks [24]. A second design detail, *dormant-covariance refresh*, keeps the inverse-covariance P_d of every readout current with the feature stream (weights frozen, P updated with zero error). Section 4.3 measures it in isolation: advancing the inverse covariances while the readout weights stay frozen learns nothing, so the refresh is not what carries the routing result.

3.5. M5: state-norm homeostat

M5 is a closed-loop controller on the whitened-state norm: the effective time step is adjusted so that the norm is held in a band around its target $\sqrt{N}$ (the whitening’s unit-variance intent),

$$\Delta_t = \text{clip}(\Delta_{t-1} + \eta(\|h_t^w\| - \sqrt{N}), 1, \Delta_{\max}), \quad h_t = A^{\Delta_t} h_{t-1} + B e_{t-1}, \quad (6)$$

with $\eta = 0.05$, $\Delta_{\max} = 10$. M5 is a feedback regulator, not a fixed normalization: it responds to the measured state and bounds the slow-channel

accumulation that otherwise makes the full-state EMA a non-stationary statistic. M5 is evaluated as a mechanism on the two-domain task (Section 4.3) and is *not* part of the benchmark stack (Section 4.4).

3.6. Tasks and discipline

Two-domain protocol (s18): two biased-bigram Markov domains (shift-1 / shift-7 with disjoint bias sets), alternating every 3000 tokens (known switch instants), 6 segments, 18k tokens. **Four-domain benchmark**: four domains (shifts 1/3/7/11, disjoint 4-symbol bias sets) on a cycling schedule with *irregular* switch intervals (Uniform 2500–3500), 8 segments, ~ 24 k tokens. All seed rules and metrics (stream perplexity, forgetting perplexity on held-out previous domain, post-switch adaptation time) follow [6] verbatim. Every comparison is paired per seed over 10 seeds with sign consistency ($n_{\text{pos}}/10$); the Transformer+LoRA reference is the tiny online-LoRA model of [6] retrained on each task. Two presentation conventions apply throughout: Figs. 1 and 3 show per-arm means with ± 1 standard-deviation error bars over the 10 seeds, Fig. 2 shows means only, and per-seed paired differences are summarized by sign consistency (and, where quoted, paired t values) rather than tabulated with confidence intervals — the sign-consistency counts, not the t values, are the robustness statistics, since extreme t values are sensitive to single seeds.

4. Results

4.1. P1/P3a: the readout needs the input path

Fig. 1 shows stream perplexity on the two-domain protocol (10 seeds) for the state-based readout arms and the input-path control, against the Transformer+LoRA reference (TF-A1, 15.01; TF prefixes the Transformer-host arms throughout). The stage labels P1–P4 — and P3a, the readout-arm sub-experiment — are this paper’s own development-stage labels (set in the order the arms were built and falsified; the companions use their own stage names and no shared label sequence is implied). Every state-feature readout — log-normal spectrum (LN-raw 114.99, LN-whiten 115.87), coverage spectrum (CV-whiten 81.31), with the current-token projection added (CV-skip 58.47) — is worse than the reference on every seed (0/10) under the reported clipped-linear metric. The input-path control ($\phi = [Be_{t-1}; 1]$, no state) reaches 11.75, better than the reference on 10/10 seeds (-3.26). Proposition 4.1 shows why exact recovery is impossible; the corollary below scopes the measured failure. The ranking under this metric is a *calibration* ranking (Section 4.7):

coverage-spectrum arms park 10–12% of tokens on the clip floor and would look *better* than the input path on unclipped CE, so the design conclusion — consume the additive input path — is stated as a requirement for calibrated next-token scores under the squared-loss readout, not as a claim that the state lacks linear information.

Proposition 4.1 (Deconvolution impossibility for the linear state readout) *Let the host be $h_t = \Lambda h_{t-1} + B e_{t-1}$ with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_N)$, $0 < \lambda_i < 1$ distinct, and $B \in \mathbb{R}^{N \times V}$ with i.i.d. continuous entries. For almost every such (Λ, B) , no affine-linear readout $\hat{e}_t = U h_t + b$ recovers the most recent input exactly ($\hat{e}_t = e_{t-1}$ for all token histories).*

Here Λ is the host matrix A of Eq. (1) with entries $\lambda_i = e^{-1/\tau_i}$; we write Λ inside the proposition to avoid collision with the RLS forgetting factor λ of Section 3.2.

Proof. Unrolling the recurrence gives the mixture $h_t = \sum_{j \geq 0} \Lambda^j B e_{t-1-j}$ (with $e_{-1} = 0$). Exact recovery for every history requires $U B e_{t-1} = e_{t-1}$ for every token, i.e. $U B = I_V$, and $U \Lambda^j B = 0$ for every $j \geq 1$. The annihilation conditions say U vanishes on $S = \text{span}\{\Lambda^j B u : j \geq 1, u \in \mathbb{R}^V\}$. Because the λ_i are distinct and B is random, the Krylov family $\{\Lambda^j B\}_{j \geq 0}$ generates $\mathbb{R}^N$ almost surely (the N vectors $\Lambda^j b_\ell$ for a single random column b_ℓ form a Vandermonde system with distinct nodes, hence are linearly independent a.s.), so $S = \Lambda \text{span}\{\Lambda^j B\}_{j \geq 0} = \Lambda \mathbb{R}^N = \mathbb{R}^N$ (Λ is invertible). Hence $U = 0$, contradicting $U B = I_V$. Whitening replaces h_t by $D h_t$ with a positive diagonal D — the elementwise whitening scale of Eq. (2), $\sqrt{N(1 - \lambda_i^2)}$ per channel — i.e. $U' \mapsto U' D$; the argument is unchanged, so the contamination is structural, not a scale artifact. $\square$

The empirical corollary of Proposition 4.1 is the measured 0/10 failure of every state-feature readout under the clipped-linear stream metric: a pooled linear readout cannot turn the state mixture into calibrated next-token scores, and the input-path feature $\phi = [B e_{t-1}; 1]$ enters the state equation as the $j=0$ term — exactly the component the readout must consume. Coverage-spectrum arms are not information-theoretically empty (Section 4.7): they achieve lower unclipped CE than the input-path control while clipping far more often. The proposition bounds *exact* recovery; a boundary analysis (data/s35_readout_boundary_probe_v1.csv) shows the measured failure is a property of the pooled readout under this metric rather than an absence of linear information. The probe is a linear decoder fit by least squares on the first half of each stream (mapping the whitened channels to the previous-token

identity) and evaluated out-of-sample on the second half by argmax agreement; chance is the uniform 32-symbol vocabulary level ($1/32 \simeq 3.1\%$), and the quoted ranges span per-channel, per-seed accuracies. The most recent input token (e_{t-1} ; called "the current token" at prediction time) is linearly decodable by argmax readout from the fast channels ($\tau_i \leq 8$) of the whitened state at 88.9–99.7% out-of-sample accuracy (chance 3.1%; 95.4–99.9% with $\tau_i \leq 30$), whereas the slow channels ($\tau_i \geq 100$) decode at chance (0.034–0.049): the older-token contamination is a slow-channel property, and whitening rescales it without removing it.

The in-sample ridge oracle is itself window-dependent (28.8–33.4 on the full window vs. 16.2–18.4 on the first half) and non-monotone in the feature set (adding the clean projection makes it worse on every seed: 16.6–19.9 vs. 7.15–7.36 for the projection alone) — artifacts of evaluating clipped linear outputs by perplexity, not bounds on recoverable information. Restricting the readout to the fast channels does not repair the next-token readout either: fast-channel-only linear fits and two-stage decode-then-table compositions still fail out-of-sample (test ppl 68–104 vs. 13.9–17.4 for the pooled transition table fit on one half and evaluated on the other) — the token is linearly present, but the closed-form squared-loss readout never turns it into calibrated next-token probabilities, which is exactly what the input-path feature achieves by encoding the token itself. Within the RLS-only constraint the honest resolution remains the additive input path.

Fixed Mamba-style multiplicative gates ($\hat{y} = W(h^w \odot \sigma(\gamma C B e_{t-1}))$, C fixed random, $\gamma \in \{1, 5\}$) do not repair the state readout: stream perplexity 77.19 ($\gamma=1$) / 72.21 ($\gamma=5$); the sharper gate does not repair it either (0/10 vs. the reference). Within the tested two-point $\gamma \in \{1, 5\}$ sampling the negative therefore does not appear to be a tuning artifact; a wider gate-parameter grid was not swept, so we state this design boundary only for the tested settings. Selectivity in real Mamba-style hosts is *learned* end-to-end; a fixed random gate carries the previous token only multiplicatively inside a time-varying state, which is noise, not signal. Within our RLS-only constraint the honest resolution is the additive input path — which is what Mamba’s input branch provides. We flag the learned-selectivity route as the natural extension rather than crossing the RLS-only boundary in this work.

4.2. P2: routing transfers, gating is readout-dependent

Table 1 and Fig. 2 (left). The three host arms are A1 (bare: the input-path RLS readout only), A2 (the pause-learning policy), and A3 (hysteresis-gated

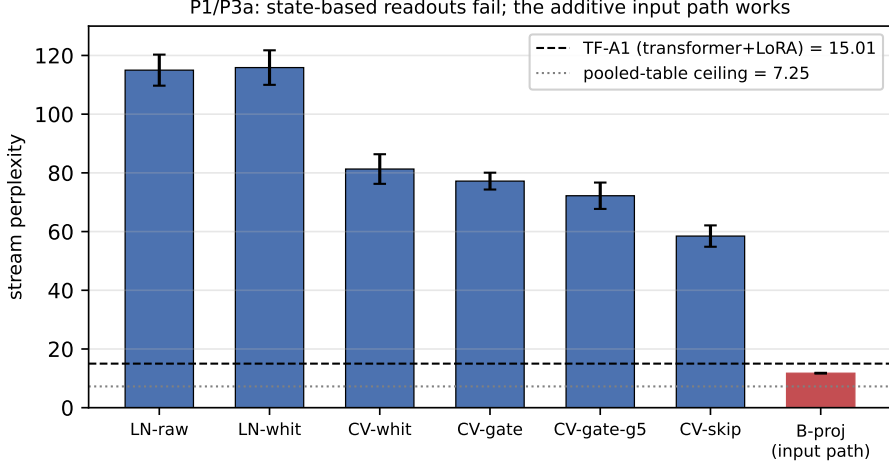


Figure 1: P1/P3a (10 seeds): stream perplexity of state-based readout arms vs. the input-path control on the two-domain streaming task, under the clipped-linear metric of Section 3.2. All state-feature readouts fail (0/10 vs. the Transformer+LoRA reference TF-A1, dashed); fixed gates do not help; the additive input path (Be_{t-1}) reaches 11.75, below TF-A1, approaching the in-sample pooled-table ceiling 7.25 (mean over seeds; range 7.15–7.36; the same table fit on one half and evaluated on the other transfers at ~ 15 across the domain-mix shift, so the input path also beats the non-adapting static table). The dotted line drawn inside the figure marks this same 10-seed mean ceiling 7.25. This ranking is floor-sensitive: see Section 4.7.

hard routing between two domain specialists — a discrete nearest-reference selector, not the soft weighting of M4, Eq. 5). M3 (fast-channel EMA) supplies the domain statistic the gate reads: over the 5 known switches of the two-domain protocol it detects a mean of 2.80/5 at $\tau_m=200$, falls to 0.30/5 at 500, and detects none at 1000 and 2000 (per-seed counts in `switches_detected`, `data/s20_ssm_m3_routing_v1.csv`) — the detector does not track the switching, yet the gate is still useful. Routing (A3) with two domain specialists improves forgetting at $\tau_m \leq 1000$ under the reported metric (-2.05 , -1.87 , -1.20 ppl; 10/10 seeds) and is neutral at 2000 (-0.01 , 5/10; the per-seed paired-difference mean is -0.007 , rounded to the table’s two-decimal convention in Table 1). The 2000 neutrality is itself floor-sensitive: under both unclipped scorings the $\tau_m=2000$ forgetting gain is robust (-1.28 per-arm unclipped / -1.07 same-token-set, 10/10; Section 4.7) — the clip floor masks this gain rather than creating it. The forgetting gain is largest where the detector still fires: seeds with at least one detection gain -1.99 ppl

on forgetting against -0.98 for seeds with none. The detector is therefore not what makes the routing work — the gain survives at $\tau_m=1000$, where it never fires — but it marks where the gain is strongest under the reported metric. A3’s stream cost in the frozen-covariance variant ($+0.38$ to $+4.08$) disappears in the M4 variant that routes softly *and* trains the specialists online, which changes the routing rule and the specialist update together (Section 3.4, Section 4.3).

The pause-learning arm (A2 — the paused-readout policy, distinct from the fixed multiplicative feature gates of Section 4.1) appears to improve stream perplexity at every tested τ_m under the reported clipped-linear metric (-1.52 to -2.45 , 10/10), the opposite of the companion’s A2 falsification (0/10). That stream margin does not survive floor removal: A2 itself clips on only $\sim 4 \times 10^{-5}$ of stream tokens, while A1 clips on 1.58%, so the reported A2-over-A1 gain is largely A1’s floor penalty. Under the unclipped metric A2 is *worse* than A1 on stream at every τ_m ($+1.45$ to $+2.38$, 0/10). Forgetting — already printed in Table 1 — is worse than A1 at $\tau_m \in \{200, 500\}$ ($+0.52$, 0/10; $+0.03$, 5/10) and only weakly better at $\tau_m \in \{1000, 2000\}$ (-0.23 , 10/10; -0.37 , 9/10); under the unclipped metric those large- τ_m gains shrink further (-0.04 , 8/10; -0.18 , 9/10). By the paper’s own rubric — a pause policy must not buy stream by sacrificing retention — A2 does not pass. Two honest comparisons with the companion remain: the companion’s A2 paused LoRA plasticity while keeping the head active, whereas here pausing the only trainable component pauses all learning; and the companion’s 0/10 falsification used a fast Adam/LoRA readout whose within-domain knowledge decays when paused, while this RLS readout stays near the pooled-table optimum. The pause policy’s effect is therefore readout-dynamics-dependent *and*, on this host, metric-dependent. We report this qualification rather than editing the companion paper.

4.3. P3: specialist retention at the tested point, and state regulation is conditional

Fig. 2 (right): at $\tau_m=500$, soft routing (M4, Eq. 5) improves forgetting over the bare single readout A1 (-1.99 ppl, 10/10), while abrupt hard switching retains purer specialists than soft routing on forgetting (-2.37 vs. -1.99 vs. A1; soft – abrupt = $+0.38$ ppl, 1/10). Under the reported clipped-linear metric soft also appears to beat abrupt on *stream* (-1.81 ppl, 10/10); that margin reverses under the unclipped metric ($+0.76$, 0/10; Section 4.7). We therefore do not claim “gentle wins” on stream. Soft routing with both

arm	τ_m	stream	Δ_{ppl}	forget	Δ_{forget}
A1 bare	—	11.75	—	8.39	—
A2 pause	200	10.24	−1.52 (10/10)	8.91	+0.52 (0/10)
A2 pause	500	9.53	−2.22 (10/10)	8.42	+0.03 (5/10)
A2 pause	1000	9.30	−2.45 (10/10)	8.15	−0.23 (10/10)
A2 pause	2000	9.70	−2.05 (10/10)	8.02	−0.37 (9/10)
A3 route	200	12.14	+0.38 (0/10)	6.33	−2.05 (10/10)
A3 route	500	13.30	+1.55 (0/10)	6.52	−1.87 (10/10)
A3 route	1000	14.81	+3.06 (0/10)	7.19	−1.20 (10/10)
A3 route	2000	15.83	+4.08 (0/10)	8.38	−0.01 (5/10)

Table 1: P2 (10 seeds, two-domain protocol): routing (A3) retains domain specialists (forgetting improves at $\tau_m \leq 1000$, 10/10) at a frozen-covariance stream cost that disappears in the M4 variant which routes softly and trains the specialists online (Section 4.3). Pause-learning (A2 — the paused-readout policy, not the multiplicative feature gates of Section 4.1) shows a reported stream gain that reverses under the unclipped metric and a forgetting cost at small τ_m (Section 4.7).

specialists trained online improves the reported stream result from the +1.55 of the hysteresis routing in Table 1, whose inactive readout is fully frozen, to −3.54 against the bare host (10/10) — also floor-sensitive, since A1 clips more than A3-soft (1.58% vs. 0.28%). Because that comparison changes the routing rule and the training of the specialists at once, we isolate the second factor (data/s37_dormant_p_probe_v1.json): with the same soft routing but the readout *weights frozen* and only the inverse covariances refreshed, stream and forgetting perplexity are both exactly 32.0 — chance for the 32-symbol vocabulary — on all 10 seeds, i.e. worse than the bare host by +20.2 (0/10). Refreshing the covariance learns nothing; the entire gain comes from training the specialists.

M5 (Eq. 6) bounds the whitened-state norm (mean 11.3 in band vs. 50.2 for the bare host, whose norm drifts upward on the slow timescales — a long transient toward a large but bounded steady state under the switched, non-stationary stream, not unbounded growth; the host is contractive), restores the *full*-state EMA as a valid domain statistic (5/5 switch detections vs. 0.6/5 — a switch counts as detected when the hysteresis nearest-reference estimate on the metadata statistic flips to the incoming domain at the known switch instant, and 0.6/5 averages the per-seed flip counts over the bare host — the non-stationarity of Section 3.3 fixed at the source), and recovers from a

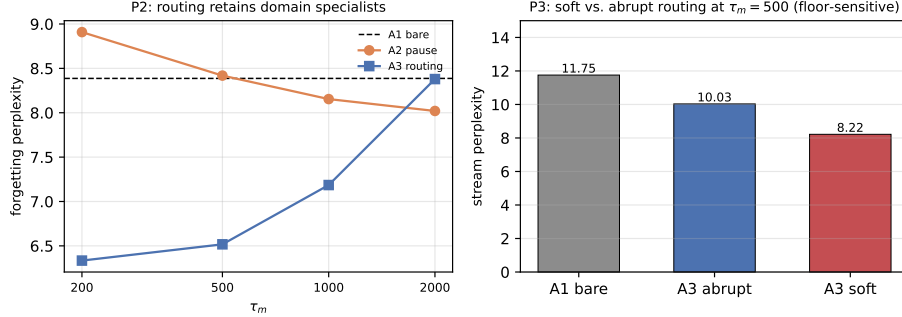


Figure 2: Left (P2): forgetting perplexity vs. τ_m — routing (A3) retains domain specialists below the bare baseline (A1, dashed) at every tested τ_m ; pause-learning (A2 — the paused-readout policy of Section 4.2, not the multiplicative feature gates of Section 4.1) does not at $\tau_m \leq 500$ and crosses below the baseline only at the two slowest metadata timescales. Right (P3): soft routing shows a lower reported stream perplexity than abrupt at the tested $\tau_m=500$ point, but that margin reverses under the unclipped metric (Section 4.7); abrupt switching retains purer specialists on forgetting.

magnitude-10 state kick in ~ 10 tokens (Δ_t settles near 5.4; the disturbance is a single-step additive kick of magnitude 10 in a random unit direction applied to the state at the first switch, and recovery is measured as the first step at which the whitened-state norm returns to within the $1.05\sqrt{N}$ band (the band’s upper edge)). This is mechanism-level evidence: on this architecture the readout is decoupled from the state (input path), so a direct “stream performance under disturbance” test is not meaningful here. These two-domain results do not transfer to the four-domain benchmark, where M5 is harmful (Section 4.4).

4.4. P4: benchmark on four-domain irregular switching

Fig. 3 and Table 2. All P4 runs — the four-domain benchmark, the real-text transfer, and the follow-up comparisons below — use the $\tau_m=500$ selected by the P2 sweep (with the data-adaptive κ of Eq. (5)), fixed thereafter and not re-swept. The four-domain irregular-switch protocol is a deliberate upgrade over the two-domain stage tasks (four domains, irregular switch intervals, a longer stream of $\sim 24k$ vs $18k$ tokens): a more demanding generalization test than the protocol on which the stage experiments were run. On this protocol, the full stack (M1 input-path readout + M3 metadata + M4 soft routing) beats the bare host robustly on *forgetting* (-4.47 reported, -4.53 unclipped; 10/10 both). The reported stream margin (-2.25 , 10/10) reverses under the

unclipped metric (+1.45, 0/10) because SSM-bare clips on 1.16% of stream tokens while REDEM clips on 0.07%; we therefore treat the P4 stream win as floor-sensitive and lead with retention. Against the Transformer+LoRA reference the full stack wins on both axes under the reported metric (stream -9.28 , forgetting -10.08 ; 10/10); the unclipped metric is undefined for TF-A1 (softmax CE, no clip floor). Even the bare host beats the transformer reference 10/10 on reported stream (-7.03): the RLS readout’s second-order convergence dominates online Adam [25]/LoRA when domains switch faster than the LoRA can track (the reference reaches only 22.46 ppl, below the uniform-chance 32).

Real-text transfer. The same protocol on two public-domain books (Alice’s Adventures in Wonderland vs. A Tale of Two Cities, character -level, case-folded, 32-symbol vocabulary) reproduces the result: REDEM-SSM beats SSM-bare robustly on forgetting (-0.46 reported, 9/10; -0.33 unclipped, 9/10). The reported stream margin (-1.26 , 10/10) nearly vanishes under the unclipped metric (-0.05 , 7/10). Against TF-A1 the full stack wins on both reported axes (stream -5.23 , forgetting -2.01 ; 10/10); even the bare host beats TF-A1 10/10. The fast-channel metadata separates the two sources, and soft routing retains per-source specialists, although the forgetting gain is smaller than on the synthetic task — the two books share English letter bigram statistics, so their specialists overlap more than the four synthetic domains’. The first-order readout reaches the char-bigram regime (ppl ~ 12 – 13); higher-order text structure is out of its reach and is reported as a boundary, not a result. A count-based bigram oracle pins this boundary quantitatively (data/s31_char_bigram_oracle_v1.csv): the true per-source first-order ceiling, estimated from the full books and evaluated on the same per-seed windows, is ppl 10.97 ± 0.18 (stream) — REDEM-SSM’s 12.07 sits within ~ 1.1 ppl of its structural ceiling (the residual is finite-sample loss of the streaming readout), and the Transformer reference (17.31) fails to reach even this first-order ceiling, so its deficit is not about higher-order structure. One caveat on the ceiling: it is fitted on the full books and evaluated on the same per-seed windows, so it is in-sample for the evaluation window and optimistic; a ref-window-fitted variant of the same oracle exists in the released oracle data and is not quoted here, so the “within ~ 1.1 ppl of ceiling” statement should be read with that convention in mind.

Reference fairness and mechanism transfer (follow-up). Two legitimate objections to the TF-A1 comparison are that the reference uses the untuned s18 hyperparameters and that it carries no routing mechanism. We close both on

the same four-domain protocol (10 seeds; data/s26_ssm_p4_fair_tf_v1.csv). (i) A small tuning grid (four learning rates from 3×10^{-4} to 10^{-2} ; ranks 16 and 32) improves the bare reference’s stream perplexity to 14.86 at $\text{lr}=3 \times 10^{-3}$, rank 32 — still worse than REDEM-SSM on 10/10 seeds (-1.68 , paired $t = -8.8$; the best config’s forgetting collapses to 62.2 vs. 8.93, so the tuning that helps the stream destroys retention and the default reference is the balanced representative). The P4 margin is therefore robust to reasonable reference tuning, though smaller on stream. (ii) The s18 routing mechanism (A3) generalized to four adapters (TF-A3, $\tau_m=500$) recovers the specialists: forgetting drops from 19.01 to 10.84 (-8.17 , 10/10) — slow-trace routing transfers to the Transformer host on this second protocol (Paper C) — but does not repair the stream: TF-A3 stays at 21.77, losing to REDEM-SSM on 10/10 seeds ($+8.59$). The stream advantage of the input-path RLS readout is a property of the stateful host, not of the routing mechanism: the mechanisms transfer across the two hosts tested (with unequally tuned reference baselines on the two hosts), and we do not claim host-agnostic transfer beyond them — the stream performance is not host-agnostic.

M5 in the benchmark (negative). The P3 state-norm homeostat was deliberately excluded from the P4 stack; adding it confirms the decision empirically (data/s33_ssm_p4_m5_v1.csv): REDEM-SSM+M5 is *worse* on all 10 seeds (stream $+1.53$, forgetting $+2.90$; paired $t = +22.8/+26.5$ on the same ordering). A plausible mechanism — offered as an interpretation, not a directly measured result — is that the Δ_t modulation departs from the per-channel whitening calibrated at $\Delta_t = 1$, so the readout’s features lose their calibrated scale on a stream that is already well-conditioned (the norm never runs away — mean $\Delta_t \approx 5.3$ means the controller is constantly active). We have not quantified the per-channel variance mismatch at $\Delta_t \approx 5.3$, nor run an M5-plus-recalibrated-whitening control, so this attribution is a hypothesis; the empirical negative itself (all 10 seeds) is the solid part of the result. M5 remains a mechanism-level safeguard for the spike regime (P3, E2), not a benchmark component.

4.5. P4 factor ablation: what M3 and what M4 contribute

Table 2’s endpoints (M1 vs. M1+M3+M4, with M4 the soft-routing instantiation of the Introduction’s numbering note) do not separate the benchmark roles of the fast-channel metadata (M3) and of soft routing (M4). The factor ablation (data/s38_ssm_p4_m3m4_ablation_v1.csv, same seeds, protocol, and $\tau_m=500$ as Table 2) adds two intermediate arms: *uniform*, four

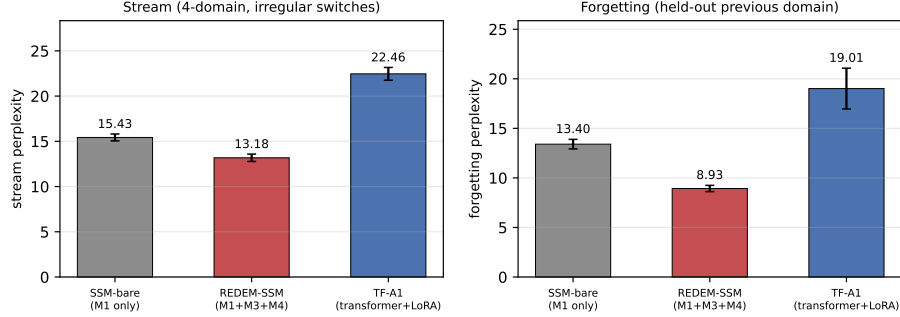


Figure 3: P4 (10 seeds, four-domain irregular-switch protocol): the full REDEM-SSM stack beats the bare host robustly on forgetting and the Transformer+LoRA reference on both reported axes (paired, 10/10; stream vs. bare is floor-sensitive, Section 4.7).

specialists trained with fixed soft weights 1/4 (M4 structure without M3 assignment), and *hard*, hard nearest-reference assignment on the M3 EMA with the inactive specialist frozen (M3 assignment without soft mixtures).

Reading Table 3: (i) specialists without assignment (uniform) are close to the bare host on stream (+0.46, 2/10) and only weakly better on forgetting (−0.43, 9/10) — M4 structure alone does not carry the benchmark; (ii) hard M3 assignment is what retains specialists on forgetting (−5.24, 10/10) but pays on stream (+2.90, 0/10); (iii) soft M4 recovers the stream margin relative to hard routing (−5.14, 10/10) while remaining slightly worse than hard on forgetting (+0.77; hard better on 8/10). The full stack therefore works because M3 assignment provides domain memory and soft M4 turns that memory into a stream-usable mixture — not because either piece alone is sufficient.

4.6. Classical ESN+RLS baseline under the same protocol

A separately wired random reservoir (ESN) with an online RLS readout on the reservoir state is the classical fixed-substrate-plus-linear-readout reference (Jaeger’s echo-state network). We ran it on the identical P4 stream, seeds, and metrics (data/s40_esn_rls_p4_baseline_v1.csv): $r_t = (1 - a)r_{t-1} + a \tanh(W_{\text{in}}e_{t-1} + W_{\text{res}}r_{t-1} + b)$ with sparse W_{res} scaled to spectral radius ρ , $a = 0.5$, input scaling 1, and the same RLS recurrence as the SSM arms.

The classical ESN is far worse than the bare diagonal host with the input-path readout (stream 98.8 vs. 15.4; paired ESN-128 – bare = +83.4, 0/10) and than the full stack (+85.6, 0/10). Wider reservoirs and edge-of-stability

arm	stream ppl	forgetting ppl
SSM-bare (M1 only)	15.43	13.40
REDEM-SSM (M1+M3+M4)	13.18	8.93
M4 = soft routing over specialist readouts; M3 = fast-channel metadata (Introduction)		
TF-A1 (transformer+LoRA)	22.46	19.01
paired (10/10): REDEM-bare $-2.25/-4.47$		
stream floor-sensitive (unclipped $+1.45/-4.53$); unclipped undefined for TF		
REDEM-TF $-9.28/-10.08$; bare-TF $-7.03/-5.61$		
M3/M4 factor split and classical ESN control: Tables 3–4		

Table 2: P4 (10 seeds, four-domain irregular-switch protocol): the full stack (input-path RLS readout, fast-channel metadata, soft routing over four specialists) beats the bare host robustly on forgetting and the Transformer+LoRA reference on every seed under the reported metric. The stream margin against the bare host is floor-sensitive (Section 4.7). The transformer reference uses the companion’s hyperparameters, untuned for four domains.

ρ do not close the gap. This is the same failure mode as the P1 state-mixture readout: a linear readout on a decaying reservoir state mixture does not calibrate next-token scores on this protocol. The bare M1 arm is not a stand-in for a classical ESN — the input path is a different topology — but a classical ESN does not rescue the host either. The one-hot control (no reservoir, 12.4) sits below the reported bare-host stream (15.4) only because that bare value is clip-inflated; on the unclipped metric the bare host (≈ 11.5) is the better linear arm (Section 4.7).

4.7. Metric sensitivity of the clipped-linear perplexity

The squared-loss readout is scored by $\text{CE} = -\ln \text{clip}(\hat{y}_t[e_t], 10^{-12}, 1)$ (Section 3.2). Each clipped token contributes a fixed $-\ln(10^{-12}) \simeq 27.63$ nats. Table 5 compares every headline paired difference under three scorings: (i) the *reported* clipped-linear metric; (ii) the *per-arm unclipped* sensitivity metric ($\text{ppl}_{\text{unclipped}}$, Section 3.2; each arm conditioned on its own $\hat{y} > 0$ tokens); (iii) the *same-token-set* metric — mean CE of both arms restricted to tokens on which *both* arms have $\hat{y} > 0$, which removes the selection bias of (ii). Lower is better; negative Δ favors the first arm. Same-token-set values are the exponential of the mean common-set CE (they are not the arithmetic mean of per-seed perplexities). Per-token arrays for (iii) are written to `data/per_token/` by the s19–s23 and s33 scripts. Three facts structure the reading:

arm	stream ppl	forgetting ppl
SSM-bare (M1)	15.43	13.40
SSM-uniform (M4 without M3)	15.89	12.98
SSM-hard (M3 without soft M4)	18.32	8.16
SSM-REDEM (M1+M3+M4 soft)	13.18	8.93
uniform – bare: +0.46/−0.43 (2/10, 9/10)		
hard – bare: +2.90/−5.24 (0/10, 10/10)		
REDEM – bare: −2.25/−4.47 (10/10, 10/10)		
REDEM – hard: −5.14/+0.77 (10/10, 2/10)		

Table 3: P4 M3/M4 factor ablation (10 seeds, same protocol as Table 2). Uniform specialists alone barely move either axis; hard M3 assignment buys forgetting at a stream cost; soft M4 recovers stream relative to hard routing while giving up a small forgetting margin. Reported metric; unclipped values in the data file.

arm	stream ppl	forgetting ppl
ESN-lin (one-hot control)	12.43	13.41
ESN-128 ($\rho = 0.9$)	98.78	97.76
ESN-128 ($\rho = 1.0$)	99.69	95.30
ESN-256 ($\rho = 0.9$)	97.10	87.83
SSM-bare (Table 2)	15.43	13.40
SSM-REDEM (Table 2)	13.18	8.93

Table 4: Classical ESN+online-RLS on the P4 protocol (10 seeds). Separately wired reservoirs fail out of sample on both axes (0/10 vs. the one-hot control and vs. SSM-bare/SSM-REDEM, seed-aligned).

1. **Stream rankings that reverse.** Soft vs. abrupt routing, A2 pause vs. A1, and REDEM-SSM vs. SSM-bare reverse sign under both unclipped scorings (10/10 reported, 0/10 under (ii) and (iii)). Same-token-set magnitudes are milder than per-arm unclipped (+0.38 vs. +0.76 ppl for soft – abrupt stream), as expected once the floor tokens are removed from both sides. The real-text stream margin is nearly zero under both unclipped scorings. These stream claims are floor-driven and are not primary conclusions.
2. **Forgetting rankings that hold.** REDEM vs. bare on P4 (same-token-set −4.44 ppl, 10/10), real-text (−0.26, 10/10), A3 routing vs. A1 (same-token-set −2.74 / −2.47 / −1.94 / −1.07 ppl at $\tau_m \in$

$\{200, 500, 1000, 2000\}$, 10/10 at every tested τ_m , including the reported-neutral 2000 point), and the M5 negative result all survive every scoring. A3’s holdout clip fraction is only 0.55–0.73%, contributing ~ 9 –11% of its reported holdout cross-entropy — the forgetting win is obtained despite a mild floor penalty, not because of it.

3. **Calibration, not linear unavailability.** Coverage-spectrum state arms beat the input-path control B-proj on unclipped stream CE under both (ii) and (iii) — same-token-set CV-gate – B-proj = -1.57 ppl (10/10) — while parking 10–12% of tokens on the floor; log-normal arms fail under every scoring. The CV arms carry more features than the input-path control (whitened state and/or the current-token projection vs. $[Be_{t-1}; 1]$ alone), so this unclipped advantage reflects both residual state information and feature-count asymmetry; it is a calibration diagnostic, not a controlled information comparison. The input path is required because it is *calibrated* under this evaluation, not because the state lacks linear information (Section 4.1). On the same common set, A3 routing also beats A1 on stream at $\tau_m \leq 1000$ (10/10), which the reported metric masks with A3’s higher clip rate.

The unclipped metrics are undefined for TF-A1 (softmax cross-entropy, no clip floor). Per-arm unclipped values remain diagnostics rather than a replacement main table; the same-token-set column is the fair paired comparison on a common support, but it still excludes the tokens on which at least one arm scores non-positive — those tokens are exactly where calibration fails, so no single unclipped summary replaces the reported metric plus the clip fractions.

5. Epistemic scope: knowability, calibration, and floors

This section fixes the boundary of Proposition 4.1 and of the triple scoring so that the host requirements can serve as a *diagnosis* layer rather than as an overclaim. Three distinctions are load-bearing and are kept separate throughout: (i) what a function class can recover from the state; (ii) whether a particular head is *calibrated* under a particular instrument; (iii) which paired rankings survive once the instrument’s floor is removed.

5.1. Extension of Proposition 1 to nonlinear readouts

Proposition 4.1 is an impossibility result for *affine-linear* maps under *exact* recovery of e_{t-1} from the linearly-decayed mixture. It does not constrain

paired comparison (Δ ppl)	reported	unclipped	same-set
soft – abrupt, stream (s21)	−1.81 (10/10)	+0.76 (0/10)	+0.38 (0/10)
soft – abrupt, forget (s21)	+0.38 (1/10)	+0.78 (0/10)	+0.71 (0/10)
A2 – A1, stream $\tau_m=500$	−2.22 (10/10)	+1.68 (0/10)	+1.38 (0/10)
A2 – A1, forget $\tau_m=500$	+0.03 (5/10)	+0.22 (0/10)	+0.23 (0/10)
A3 – A1, stream $\tau_m=200$	+0.38 (0/10)	−2.16 (10/10)	−1.78 (10/10)
A3 – A1, forget $\tau_m=200$	−2.05 (10/10)	−2.98 (10/10)	−2.74 (10/10)
A3 – A1, forget $\tau_m=500$	−1.87 (10/10)	−2.68 (10/10)	−2.47 (10/10)
A3 – A1, forget $\tau_m=1000$	−1.20 (10/10)	−2.16 (10/10)	−1.94 (10/10)
A3 – A1, forget $\tau_m=2000$	−0.01 (5/10)	−1.28 (10/10)	−1.07 (10/10)
REDEM – bare, stream (s22)	−2.25 (10/10)	+1.45 (0/10)	+1.10 (0/10)
REDEM – bare, forget (s22)	−4.47 (10/10)	−4.53 (10/10)	−4.44 (10/10)
REDEM – bare, stream (s23)	−1.26 (10/10)	−0.05 (7/10)	−0.21 (10/10)
REDEM – bare, forget (s23)	−0.46 (9/10)	−0.33 (9/10)	−0.26 (10/10)
M5 – REDEM, stream (s33)	+1.53 (0/10)	+1.59 (0/10)	+1.58 (0/10)
M5 – REDEM, forget (s33)	+2.90 (0/10)	+2.71 (0/10)	+2.51 (0/10)
CV-gate – B-proj, stream (s19)	+65.4 (0/10)	−3.89 (10/10)	−1.57 (10/10)
LN-raw – B-proj, stream (s19)	+103.2 (0/10)	+8.64 (0/10)	+9.82 (0/10)

Table 5: Metric sensitivity (10-seed paired means). Script labels: s19 = P1/P3a readout arms; s20 = P2 A2/A3 routing; s21 = P3 soft vs. abrupt; s22 = P4 four-domain benchmark; s23 = P4 real-text; s33 = M5 in the P4 stack. Reported: clipped-linear CE. Unclipped: each arm on its own $\hat{y} > 0$ tokens (selection-biased). Same-set: both arms on the intersection $\{\hat{y}_A > 0\} \cap \{\hat{y}_B > 0\}$ (fair paired support; still excludes tokens on which either arm clips). Unclipped columns are undefined for TF-A1. Stream rows whose sign reverses are not used as primary conclusions. Data: the archived per-experiment CSVs of the scripts listed above, plus per-token CE dumps used only for the same-token-set rescore.

nonlinear heads, and it must not be read as one. Three facts delimit a correct nonlinear extension.

(a) *Nonlinear heads do not invert the deconvolution..* A softmax (or any other continuous nonlinear) head can change the *objective* and the *scoring rule*, but it does not undo the contamination structure that Proposition 4.1 exposes: the state is still a Krylov mixture of past tokens, and the $j \geq 1$ components are still present. What the companion study [26] changes is the placement of the probabilistic map (*inside* the training objective), not a left inverse of the mixture.

(b) *The range fix is not the repair..* On the same host, stream, and seeds, post-hoc simplex projection of the same RLS scores removes every non-positive mass (neg_frac $\rightarrow 0$) and leaves stream perplexity unchanged (B-simplex 11.783 vs. B-lin-clip 11.754; Δ ppl +0.03, 0/10). By contrast, a *trained* softmax readout on the input path lowers stream cross-entropy by 0.307 nats/token (11.75 $\rightarrow$ 8.65, 10/10), at an honest same-token-set retention cost of +0.32

ppl. Under a floor-free probabilistic head the instrument ceases to inject a clip atom; the remaining question is whether the *state* is then a usable feature.

(c) *The surviving statement is a selection statement..* Under the clean softmax metric (companion C3), the ungated full whitened state is stream-neutral against the calibrated input-path baseline and retention-harmful; a 128-dimensional noise control excludes dimension/optimizer imbalance as the cause; a 32-dimensional state slice is stream-positive with mild retention cost. Hard top- k sparsity ($k = 32$) then supplies the missing selection pressure (best stream at near-baseline retention), and soft routing over two online softmax experts converts the residual stream-retention tension into a further retention gain ($9.20 \rightarrow 7.79$, 10/10). The correct nonlinear extension of the P1 diagnosis is therefore:

The pooled full state is not a reliable next-token feature even under a floor-free probabilistic head; a selected subspace is. P1 itself remains a statement about affine-linear exact recovery and is neither strengthened nor weakened by these nonlinear results.

Measurement class.. Proposition 4.1 assumes a linear measurement of the state (affine map, squared loss, clipped CE). Nonlinear measurement instruments of the state — nonlinear probes, mutual-information estimators, or other SiN-style nonlinear readouts — lie outside that assumption and may certify recoverability under a larger function class. They do *not* certify calibration of a pooled head under a squared-loss clipped score, and they do not repair the measured 0/10 failure of that head. The empirical corollary of P1 remains a fact about the pooled linear instrument: every state-feature arm fails out of sample (0/10) under the reported metric, while s35 shows the token is linearly present on the fast channels. Extending the measurement class changes what can be *known* from the state; it does not silently convert a calibration failure into an impossibility theorem, nor the reverse.

5.2. Calibration versus information: a formal distinction

Let $T_t = e_t$ be the target token, Φ_t the feature supplied to the readout, $\mathcal{G}$ a function class, and $S(\hat{y}, T)$ a scoring rule (here $S_\varepsilon(\hat{y}, T) = -\ln \max(\hat{y}[T], \varepsilon)$ with $\varepsilon = 10^{-12}$).

Information (recoverability under a class).. T is $\mathcal{G}$ -recoverable from Φ if there exists $g \in \mathcal{G}$ with $g(\Phi) = T$ almost surely. Linear recoverability takes $\mathcal{G}$ to be the affine maps; Proposition 4.1 rules out $\mathcal{G}$ -recoverability of e_{t-1} from h_t for that class. A weaker information statement is that $I(\Phi; T) > 0$, or that the Bayes risk under $\mathcal{G}$ is below chance. The s35 boundary probe establishes the weak form on the fast channels ($\tau_i \leq 8$): an out-of-sample linear decoder reaches 88.9–99.7% accuracy against chance 3.1%, while slow channels ($\tau_i \geq 100$) sit at chance (0.034–0.049). The most recent input token (e_{t-1}) is therefore *linearly present* in the state (argmax-decodable); P1 denies exact affine recovery of the mixture, not linear information on a subspace.

Calibration under an instrument.. A pair (r, S) — readout r and scoring rule S — is *calibrated on a distribution* if (i) *support validity*: $r(\Phi)$ lies in the domain of S almost surely (for clipped CE, $\hat{y}[T] > \varepsilon$ a.s., i.e. the floor does not bind), and (ii) *properness on the induced scores*: the expected score is minimized, over the allowed score range, by the true conditional distribution. Support validity is necessary for any proper interpretation of the score as a log-loss; without it, tokens on the floor contribute the constant $-\ln \varepsilon \simeq 27.63$ nats and carry no ranking information among themselves.

The two failures are not the same failure.. The measured 0/10 of every state-feature arm under the reported metric is a failure of *calibration* of the pooled squared-loss RLS head under S_ε (large clip_frac, 10–12% on coverage-spectrum arms vs. $\sim 1.6\%$ on the input path). The s35 probe is a success of *linear recoverability* on the fast channels. These coexist: high $I(\Phi_{\text{fast}}; T)$ does not imply that a closed-form squared-loss solution, scored with a clip floor, produces calibrated next-token probabilities. The forbidden inference is the one the paper already refuses to make (Section 4.1, Section 4.7): *from “the pooled readout fails 0/10” to “the state has no linear information.”* That inference is refuted by the host’s own probe.

Consequence for host requirements.. The design conclusion — consume the additive input path $\phi = [Be_{t-1}; 1]$ — is a requirement for *calibrated* next-token scores under the squared-loss RLS family, not an information-theoretic allocation of “what the state knows.” The state’s primary role remains statistical metadata (fast-channel EMA, specialist routing); residual linear information in the state is available to a selected nonlinear head (Section 5.1) but was never the object of the input-path requirement.

5.3. Floor sensitivity: formal characterization

Write the expected reported score of arm a as

$$\mathbb{E}[S_\varepsilon^{(a)}] = (1 - \alpha_a) \mu_a + \alpha_a c_\varepsilon, \quad c_\varepsilon = -\ln \varepsilon, \quad (7)$$

where $\alpha_a = \Pr(\hat{y}_a[T] \leq \varepsilon)$ is the clip fraction and μ_a is the mean score on the unclipped tokens of arm a . For a paired comparison $A - B$,

$$\Delta_{\text{rep}} = \left((1 - \alpha_A) \mu_A - (1 - \alpha_B) \mu_B \right) + (\alpha_A - \alpha_B) c_\varepsilon. \quad (8)$$

The second term is the *floor-tax differential*. A ranking is *floor-dominated* whenever $|(\alpha_A - \alpha_B) c_\varepsilon|$ exceeds the magnitude of the unclipped contrast, so that the sign of Δ_{rep} is determined by which arm parks more mass on the floor rather than by which arm ranks the non-clipped tokens better. This is exactly the reversal pattern of Table 5: A2-over-A1 stream, soft-over-abrupt stream, and REDEM-over-bare stream are favorable under the reported metric and reverse under both unclipped scorings, because the first arm in each pair clips less (or the second clips more) and c_ε is large.

Same-token-set as a partial repair.. Restricting both arms to $\{\hat{y}_A > 0\} \cap \{\hat{y}_B > 0\}$ removes the floor atoms from both sides and yields a paired comparison on a common support. It is fair for the non-clipped mass, but it silently drops the tokens on which calibration failed — the tokens that are exactly where the instrument is broken. No single unclipped summary therefore replaces the reported metric *together with* the clip fractions: the triple scoring (reported / per-arm unclipped / same-set) is the minimal disclosure that separates ranking from floor tax.

Which metric families have floors.. A scoring rule S together with a readout family $\mathcal{R}$ *carries a floor* if there exists a finite constant c and some $r \in \mathcal{R}$ such that $\Pr(S(r(\Phi), T) = c) > 0$ — an atom at a fixed penalty on a set of “invalid” or clamped outputs of positive probability. The clipped-linear family of this paper has $c = c_\varepsilon$ and empirically $\alpha \in [0.07\%, 11.3\%]$ across the measured arms of this paper (the extremes are the full stack at 0.07% and the multiplicative gate controls at 11.3%); those tokens all pay the same $\simeq 27.63$ nats. A family is *floor-free* when the output map lands in the proper domain of S almost surely and S is continuous in the scores without an absorbing constant. Softmax cross-entropy is floor-free by construction: every coordinate is strictly positive and there is no atom at a fixed penalty

(`floor_frac = 0` is structural, not a result). Large $-\ln \hat{y}[T]$ values can still occur, but they are a continuous tail, not a constant tax, and they cannot by themselves produce the sign-reversal pathology of Eq. (8). Range fixes that only relocate mass onto the floor (post-hoc simplex) inherit the floor rather than removing it [26].

Host requirements as epistemic boundaries.. The two boundaries this paper establishes are complementary and must not be collapsed into a single “the state cannot predict tokens”:

- **Knowability (“what I can know”).** Proposition 4.1 bounds exact affine recovery from the mixture; the s35 probe locates linear information on the fast channels; the nonlinear extension (Section 5.1) shows that even a floor-free head needs *selection* on the full state. Together these bound what a given function class, on a given channel subset, can recover from the host state.
- **Verifiability (“what I can verify”).** The triple scoring bounds which paired rankings a host-level claim can certify once the instrument’s floor is accounted for. Floor-dominated stream margins are not evidence of mechanism superiority; they are evidence of unequal α under a large c_ϵ .

A self-evaluating system that treats a floor-dominated ranking as capability evidence confuses the two; a reader who treats P1 as an information-theoretic void confuses exact affine recovery with recoverability. In the larger self-evaluation program, Paper D is the diagnosis layer: before any causal claim that a mechanism is responsible for a gain, the host must first admit a knowable signal and the instrument must first be able to verify a ranking. Paper C’s routing transfer and this paper’s M3/M4 ablations are causal claims about mechanisms only insofar as they clear both boundaries; the M5 negative result clears them on the four-domain benchmark and is therefore reported as robust rather than as floor-sensitive.

6. Discussion

Three evidence-based host requirements emerged, and Section 5 fixes the epistemic license of each. **(i)** A linear readout cannot produce calibrated next-token scores from a linearly-decayed state mixture under the squared-loss / clipped-linear metric: exact recovery is a deconvolution impossibility

(Proposition 4.1); coverage-spectrum state arms can look strong on unclipped CE while parking 10–12% of tokens on the clip floor, so the additive input path is required for *calibration* (Section 4.1); fixed gates do not substitute for learned selectivity. Under a floor-free softmax head the pooled full state remains stream-neutral and retention-harmful, while a selected subspace is usable (Section 5.1) — a selection statement, not a nonlinear impossibility theorem. **(ii)** The state’s role is statistical metadata: its fast channels, not the readout, carry the domain statistic (Section 4.2); on first-order tasks the slow channels are inert for the readout, and their accumulation must be masked (Section 3.3) or, where helpful, regulated (M5, Section 4.3) — the latter is conditional on the task. **(iii)** Routing retains specialists (forgetting robust under both metrics); the reported soft-over-abrupt stream margin is floor-sensitive and is not claimed; the pause-learning policy fails the retention rubric on this host once the clip floor is accounted for, so neither “gentle wins on stream” nor “the readout must remain active” is a universal principle.

The honest negative results are the paper’s spine: the naive state readout, the fixed gate, the full-state EMA metadata, the frozen-covariance routing, the refresh-only routing, and the four-domain homeostat all failed in ways that *located* the working design (input path, fast-channel metadata, soft routing over specialists trained online) rather than merely being abandoned.

Section 5 also states the program-level division of labor: Proposition 4.1 and the s35 probe bound *knowability* under a function class; the triple scoring bounds *verifiability* under an instrument. Host requirements are therefore not “the state cannot predict tokens” but a two-sided diagnosis — what the host can support, and what a prescribed metric can certify — that any later causal claim about mechanisms must clear before it is believed.

7. Limitations

CPU-only, toy-scale (state 128, $\sim 4\text{k}$ readout parameters); the input-path readout is a first-order bigram model — the real-text benchmark transfers (Section 4.4) but higher-order text structure is out of reach; linear host (no chaos; M5 is a stability-edge regulator, an analogy to the homeostat of [14, 5], not a validation of it); the learned selectivity route (real Mamba-style gating) is not crossed, and the fixed-gate negative was tested only at $\gamma \in \{1, 5\}$ (a learned-soft-gate control was not run on this host; the companion study [26] shows learned top- k selectivity does repair calibration, under the same host and protocol); the transformer reference uses the companion’s hyperparameters

untuned for four domains (the tuned reference trades stream margin for retention); the host configuration ($N = 128$, log-uniform $\tau \in [1, 3000]$, $V = 32$) is a single unswept configuration, so all results are conditional on it; τ_m was swept in the P2 stage and then fixed for all downstream tasks (no re-sweeping downstream), while κ is data-adaptive ($0.5 \times$ median reference distance) rather than fixed; per-seed paired differences are summarized by sign consistency and, where quoted, paired t values — these comparisons are exploratory in the multiplicity sense (per-claim paired tests at $n = 10$ seeds, no family-wise or false-discovery-rate correction across the paper’s comparisons), no confidence intervals or per-seed tables are given, and only Fig. 1 and 3 carry error bars (Fig. 2 does not). The classical ESN baseline (Section 4.6) is an untuned topology control ($\rho \in \{0.9, 1.0\}$, two widths), not a hyperparameter sweep of the reservoir literature; the M3/M4 factor ablation (Section 4.5) separates hard nearest-reference assignment from soft mixtures at a single τ_m . No scaling claims. Epistemic scope (Section 5) is itself limited: Proposition 4.1 is only an affine-linear exact-recovery result; the nonlinear extension rests on the companion’s softmax experiments on this same host rather than on a new impossibility theorem for continuous heads; the formal floor decomposition (Eq. (7)) assumes a two-term atom-plus-conditional structure and does not cover scoring rules with graded clamps or adaptive floors; “calibration” is used in the support-validity / proper-scoring sense of Section 5.2, not as a claim of distributional calibration curves. The knowability/verifiability split is a reporting discipline for this host family, not a general epistemology of learning systems.

8. Conclusion

REDEM mechanisms can be instantiated natively on a state-space host, and the falsifying development sequence locates the working design precisely: a calibrated input-path RLS readout, fast-channel statistical metadata, and soft routing over specialists that are trained online (a refresh of the inverse covariances alone learns nothing). A state-norm homeostat is a conditional mechanism-level safeguard — it restores full-state metadata on the two-domain task but is excluded from the four-domain stack where it hurts. Under the clipped-linear perplexity, stream comparisons against arms with a higher clip floor are artifact-sensitive and reverse once clipped tokens are excluded (Eq. (8)); forgetting gains (P4, real text, A3 routing) and the M5 negative result are robust under both metrics (Section 4.7). Proposition 4.1 is a

knowability bound for affine-linear exact recovery, not an information-theoretic void: the token is linearly present on the fast channels, and under a floor-free probabilistic head the pooled full state still fails as a next-token feature while a selected subspace does not (Section 5). The full stack (M1+M3+M4) beats the bare host on retention and the Transformer+LoRA reference on both reported axes on a four-domain irregular-switch benchmark on every seed (the reference untuned; tuned, the stream margin narrows to -1.68 while its retention collapses). On that benchmark, M3 assignment carries the forgetting gain and soft M4 recovers stream relative to hard routing (Section 4.5), while a classical ESN+RLS reservoir fails out of sample under the same protocol (Section 4.6).

The calibration prescription left open here has been executed by the companion study [26] on the same host and protocol; the numbers in the following sentences are that paper’s results, quoted with attribution and not re-derived here (this paper’s contribution is the *diagnosis* — the clip-floor artifact, the input-path requirement, the floor-sensitivity of stream rankings — while the companion contributes the *readout-form fix* and its learned selectivity): post-hoc simplex projection removes negative mass without repairing stream scores (0/10), whereas a trained softmax readout removes the clip floor and lowers stream cross-entropy by 0.307 nats/token (10/10); hard top- k sparsity then repairs the retention cost that unconstrained CE gating leaves behind, and soft routing over online softmax experts converts the residual stream-retention tension into a further retention gain. Paper D’s remaining next steps are therefore not the range fix or the objective fix — those are closed — but larger-scale and higher-order real-text evaluation, learned selectivity beyond the supplied candidate families, and host configurations beyond the single $N = 128$ log-uniform spectrum tested here. As the diagnosis layer of the program, this paper’s durable contribution is the two-sided boundary: what the host can know (Proposition 4.1, s35, nonlinear selection) and what the instrument can verify (triple scoring, floor-tax differential).

CRediT authorship contribution statement

Yaming Hu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used an AI-assisted writing and review tool in order to improve readability, check consistency and draft editorial text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Code and data availability

All simulation scripts and generated data are openly available at <https://github.com/huyamingc/REDEM> (MIT license; `requirements-lock.txt` pins the package versions used for the reported runs). The experiment-to-data mapping for the numbers reported here is given in the `data/` directory (`s19-s38` CSV/JSON pairs, one row per seed), and the companion study [26] is archived as <https://doi.org/10.5281/zenodo.22110623>.

References

- [1] A. Gu, K. Goel, C. Ré, Efficiently modeling long sequences with structured state spaces, in: International Conference on Learning Representations (ICLR), 2022.
- [2] A. Gu, T. Dao, Mamba: Linear-time sequence modeling with selective state spaces, arXiv:2312.00752 (2023).
- [3] M. Lukoševičius, H. Jaeger, Reservoir computing approaches to recurrent neural network training, Computer Science Review 3 (3) (2009) 127–149.

- [4] H. Jaeger, The “echo state” approach to analysing and training recurrent neural networks, GMD Report 148, German National Research Center for Information Technology (2001).
- [5] Y. Hu, REDEM: Online learning with meta-adaptation and structural plasticity for non-stationary environments, Zenodo (2026). Companion preprint (Paper B), <https://doi.org/10.5281/zenodo.22110606>.
- [6] Y. Hu, Dissecting online learning mechanisms: statistical memory, homeostatic recovery, and substrate physics are non-substitutable in the directions and conditions tested, Zenodo (2026). Companion preprint (Paper C), <https://doi.org/10.5281/zenodo.22110618>.
- [7] H. Jaeger, H. Haas, Harnessing nonlinearity: Predicting chaotic systems and saving energy in wireless communication, *Science* 304 (2004) 78–80.
- [8] W. Maass, T. Natschläger, H. Markram, Real-time computing without stable states: A new framework for neural computation based on perturbations, *Neural Computation* 14 (2002) 2539–2560.
- [9] A. Vaswani, N. Shazeer, N. Parmar, et al., Attention is all you need, in: *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [10] B. Liu, et al., Longhorn: State space models are amortized online learners, in: *International Conference on Learning Representations (ICLR)*, 2025.
- [11] A. Behrouz, P. Zhong, V. Mirrokni, Titans: Learning to memorize at test time, in: *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- [12] Y. Sun, X. Li, K. Dalal, J. Xu, A. Vikram, G. Zhang, Y. Dubois, X. Chen, X. Wang, S. Koyejo, T. Hashimoto, C. Guestrin, Learning to (learn at test time): RNNs with expressive hidden states, in: *International Conference on Machine Learning (ICML)*, 2025.
- [13] S. Yang, B. Wang, Y. Zhang, Y. Shen, Y. Kim, Parallelizing linear transformers with the delta rule over sequence length, in: *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. (arXiv:2406.06484).
- [14] Y. Hu, Memory and chaos in a physics-constrained relaxation substrate: phase diagram, multi-timescale forgetting, and disturbance robustness,

- Zenodo (2026). Companion preprint (Paper A), <https://doi.org/10.5281/zenodo.22109664>.
- [15] D. Verstraeten, B. Schrauwen, M. d’Haene, D. Stroobandt, An experimental unification of reservoir computing methods, *Neural Networks* 20 (2007) 391–403.
 - [16] G. Parisi, R. Kemker, J. Part, et al., Continual lifelong learning with neural networks: A review, *Neural Networks* 113 (2019) 54–71.
 - [17] M. Zinkevich, Online convex programming and generalized infinitesimal gradient ascent, in: *International Conference on Machine Learning (ICML)*, 2003, pp. 928–936.
 - [18] N. Cesa-Bianchi, G. Lugosi, *Prediction, Learning, and Games*, Cambridge University Press, New York, 2006.
 - [19] E. Hu, Y. Shen, P. Wallis, et al., LoRA: Low-rank adaptation of large language models, in: *International Conference on Learning Representations (ICLR)*, 2022.
 - [20] L. Ljung, T. Söderström, *Theory and Practice of Recursive Identification*, MIT Press, Cambridge, MA, 1983.
 - [21] S. Haykin, *Adaptive Filter Theory*, 4th ed., Prentice Hall, Upper Saddle River, NJ, 2002.
 - [22] G. C. Goodwin, K. S. Sin, *Adaptive Filtering Prediction and Control*, Prentice-Hall, Englewood Cliffs, NJ, 1984.
 - [23] N. Shazeer, A. Mirhoseini, K. Maziarz, A. Davis, Q. Le, G. Hinton, J. Dean, Outrageously large neural networks: The sparsely-gated mixture-of-experts layer, in: *International Conference on Learning Representations (ICLR)*, 2017.
 - [24] S. Hochreiter, J. Schmidhuber, Long short-term memory, *Neural Computation* 9 (1997) 1735–1780.
 - [25] D. P. Kingma, J. Ba, Adam: A method for stochastic optimization, in: *International Conference on Learning Representations (ICLR)*, 2015.

- [26] Y. Hu, Learning the readout form: objective-level calibration, sparse selectivity, and expert routing on a diagonal state-space host, companion preprint (Paper F), 2026.